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In

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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CERTIFICATE

This is to certify that “**PHYSICAL EDUCATION (SPORTS AND ATHLETICS)(BPEK559)**” sports activity is successfully carried out by **DARSHAN C M (1KT23AI015)**, a bonafide student of **Sri Krishna Institute of Technology**, in Partial Fulfillment for 6th semester Bachelor of Engineering in **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** of the VTU, Belagavi, during the year 2025-26. It is certified that all suggestions indicated for internal assessment have been incorporated in the report.

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PHYSICAL EDUCATION

INTRODUCTION TO PHYSICAL EDUCATION

Physical Education (PE) plays a crucial role in shaping the physical, mental, emotional, and social development of students. It goes beyond simple exercise routines and promotes a holistic approach to health and well-being. Through structured physical activities, fitness training, games, and athletics, students develop discipline, confidence, teamwork, and a lifelong appreciation for physical fitness.

In today's academic environment, Physical Education is recognized as an essential subject because it combats sedentary lifestyles, promotes health awareness, and reduces stress. Regular participation in physical activities is proven to improve concentration, academic performance, and emotional stability. PE also provides opportunities for leadership, cooperation, and character development, helping students learn how to handle competition respectfully and work effectively with others.

This report covers three major modules included in the syllabus:

1. **Postural deformities & Stress Management:** Postural deformities are abnormal changes in body alignment caused by poor posture, muscle imbalance, or repetitive sports activities, which can affect performance and increase injury risk. Stress management in sports involves controlling mental and physical pressure through techniques like relaxation, proper rest, and positive thinking. Both are important for maintaining overall health and improving athletic performance.
2. **Specific Games (Throwball and Long jump):** Throwball is a team sport in which players throw a ball over a net, aiming to score points by making it land on the opponent's side. It improves coordination, agility, and teamwork. Long jump is a track and field event where athletes leap forward for maximum distance after a running start, enhancing strength, speed, and balance.
3. **Aerobics:** Aerobics is a form of exercise that combines rhythmic movements with music to improve cardiovascular fitness. It helps increase stamina, flexibility, and coordination while supporting weight management. Regular practice also reduces stress and improves overall health.

The aim of this report is to emphasize the significance of Physical Education in promoting balanced growth and lifelong fitness habits. Understanding fitness principles, game skills, and athletic techniques helps students lead healthier lives and contributes to their personal, social, and physical development.

Module IV

POSTURAL DEFORMITIES

1. Introduction:

1. Posture: Alignment of body parts (head, spine, limbs)
2. Postural Deformation: Abnormal changes in body alignment
3. Common in athletes due to repetitive movements and training stress

2. Types of Postural Deformation:

2.1 Kyphosis

Kyphosis is an excessive outward curvature of the upper spine, resulting in a hunched back appearance. It is commonly seen in athletes involved in cycling or activities requiring forward bending.

2.2 Lordosis

Lordosis is an excessive inward curvature of the lower spine, often referred to as swayback. It is common in athletes who experience high stress on the lower back, such as weightlifters.

2.3 Scoliosis

Scoliosis is a sideways curvature of the spine forming an “S” or “C” shape. It can affect balance, posture, and overall body symmetry.

2.4 Flat Foot

Flat foot is a condition where the arch of the foot collapses, leading to the entire foot touching the ground. It is frequently observed in runners and athletes with inadequate footwear.

2.5 Knock Knees (Genu Valgum)

In this condition, the knees come together while the ankles remain apart. It can affect running and jumping performance.

2.6 Bow Legs (Genu Varum)

Bow legs occur when the legs curve outward, creating a gap between the knees even when the ankles are together.

3. Causes of Postural Deformation in Sports:

1. Repetitive movements in specific sports
2. Muscle imbalance (unequal strength/flexibility)
3. Poor posture during training or competition
4. Overtraining and lack of recovery
5. Improper techniques and coaching
6. Previous injuries and inadequate rehabilitation

4. Effects on Athletes:

Postural deformities can significantly impact an athlete's performance and health:

- Reduced physical performance
- Chronic pain and discomfort
- Increased risk of injuries
- Poor balance and coordination
- Long-term musculoskeletal issues

5. Risk Factors in Sports:

Certain sports have a higher risk of causing postural deformation:

- Gymnastics – spinal stress
- Weightlifting – lower back strain
- Cricket and Tennis – asymmetric movements
- Cycling – prolonged forward posture
- Swimming – shoulder and spine imbalance

6. Diagnosis:

Postural deformities are identified through:

- Physical examination by professionals
- Posture analysis techniques
- Movement and flexibility tests
- X-rays or imaging in severe cases

7. Prevention:

Preventive measures are crucial in sports:

- Proper warm-up and stretching
- Strengthening weak muscle groups
- Maintaining correct posture
- Balanced training programs
- Adequate rest and recovery
- Use of appropriate equipment

8. Correction and Treatment:

- Physiotherapy and rehabilitation exercises
- Postural correction exercises
- Yoga and flexibility training
- Use of braces (in severe conditions)
- Medical or surgical treatment (rare cases)

STRESS MANAGEMENT

1. Introduction

Stress is a natural physical and psychological response to demands or challenges. In sports, athletes frequently face stress due to competition, performance pressure, expectations, and training intensity. While a moderate level of stress can enhance performance, excessive stress can negatively affect both mental and physical health. Therefore, effective stress management is essential for athletes to perform consistently and maintain overall well-being.

2. Types of Stress in Sports

2.1 Acute Stress

Short-term stress experienced before or during competitions. It can improve alertness and performance if managed properly.

2.2 Chronic Stress

Long-term stress caused by continuous pressure, overtraining, or personal issues. It can lead to burnout and reduced performance.

2.3 Eustress (Positive Stress)

A beneficial form of stress that motivates athletes and enhances focus and energy levels.

2.4 Distress (Negative Stress)

Harmful stress that causes anxiety, fear, and poor performance.

3. Causes of Stress in Sports

- High expectations from coaches, parents, and self
- Fear of failure or losing
- Intense competition
- Overtraining and fatigue
- Injuries and recovery pressure
- Time management issues (academics + sports)
- Lack of confidence or preparation

4. Effects of Stress on Athletes

4.1 Physical Effects

- Fatigue and low energy
- Increased heart rate and sweating
- Muscle tension
- Sleep disturbances

4.2 Psychological Effects

- Anxiety and nervousness
- Lack of concentration
- Reduced confidence
- Mood swings and irritability

4.3 Performance Effects

- Poor decision-making
- Decreased coordination
- Reduced overall performance

5. Signs and Symptoms of Excessive Stress

- Constant worry or fear
- Loss of motivation
- Poor sleep patterns
- Frequent injuries
- Decreased interest in sports
- Emotional instability

6. Importance of Stress Management in Sports

- Improves focus and concentration
- Enhances performance
- Maintains mental health
- Prevents burnout
- Promotes long-term athletic development

7. Stress Management Techniques

7.1 Physical Techniques

- Regular exercise and conditioning
- Proper warm-up and cool-down
- Adequate sleep and rest
- Balanced nutrition

7.2 Psychological Techniques

- Relaxation techniques (deep breathing, meditation)
- Visualization (imagining successful performance)
- Positive self-talk
- Goal setting (short-term and long-term goals)

7.3 Time Management

- Balancing academics, training, and rest
- Planning schedules effectively

7.4 Social Support

- Support from coaches, teammates, and family
- Counseling and mental health support

8. Role of Coaches and Trainers

- Create a positive and supportive environment
- Avoid excessive pressure
- Encourage open communication
- Provide constructive feedback
- Monitor athlete's mental and physical health

9. Prevention of Stress in Athletes

- Proper training programs
- Avoid overtraining
- Regular breaks and recovery sessions
- Mental conditioning programs
- Encouraging a healthy lifestyle

Module V

SPECIFIC GAMES

THROWBALL:-

Introduction to Throwball

Throwball is a popular ball game played mainly in schools, colleges, and sports clubs. It is especially popular in India and is played between two teams separated by a net. The game requires teamwork, quick reflexes, coordination, and communication skills. Throwball is similar to volleyball, but instead of hitting the ball, players catch and throw it over the net within a short time.

The game is simple to learn and helps improve physical fitness, concentration, agility, and sportsmanship. It is widely played by both men and women as a recreational and competitive sport.

Basic Rules of Throwball

1. Each team consists of 7 players on the court.
2. The match is usually played in best of three sets.
3. A team scores a point when the opponent fails to catch or return the ball properly.
4. Players must catch the ball with both hands and throw it back within 3 seconds.
5. The ball should not touch the ground inside the team's court.
6. Players are not allowed to hold, dribble, or pass the ball among teammates.
7. The ball must be thrown over the net in a single attempt.
8. Service is done from the service area behind the end line.
9. Rotation of players occurs after winning the service from the opponent.
10. The team that first scores the required points with a minimum lead wins the set.

Court Dimensions of Throwball

Feature	Measurement
Court Length	18 meters
Court Width	9 meters
Net Height (Women)	2.2 meters
Net Height (Men)	2.3 meters
Service Area	Behind the end line
Center Line	Divides the court into two equal halves
Boundary Lines Width	5 cm

Court Description

- The throwball court is rectangular in shape.
- The court is divided into two equal halves by a net.
- Each half measures 9 m × 9 m.
- Boundary lines should be clearly marked.
- The playing surface should be flat and non-slippery for safe movement.

Importance of Throwball

- Improves physical fitness and flexibility.
- Develops teamwork and coordination.
- Enhances concentration and reflex actions.
- Encourages discipline and sportsmanship.
- Suitable for all age groups.

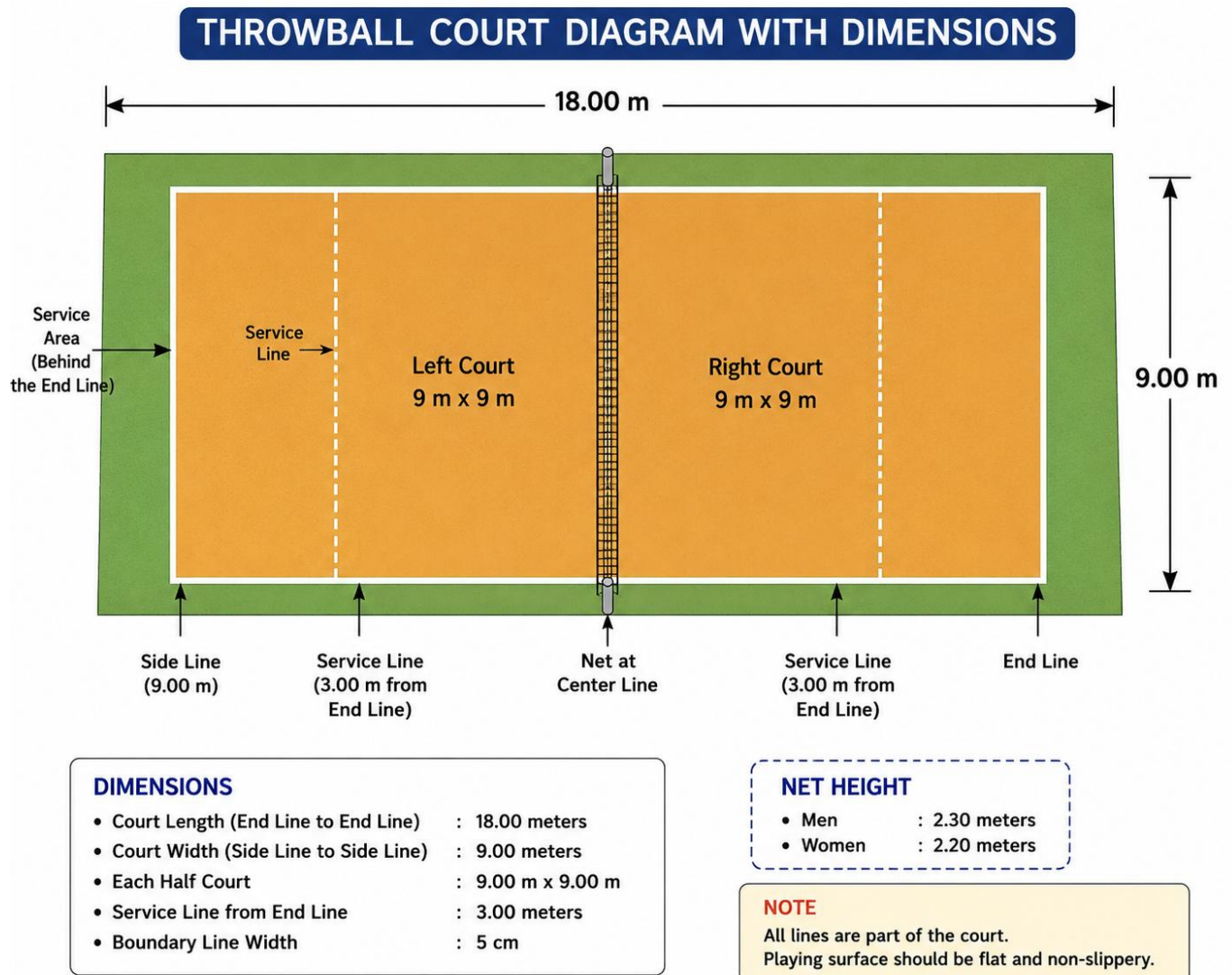


Figure 5.1 – Throw ball court



FIGURE 5.2 -PLAYING THROWBALL.

PART B ATHLETICS--LONG JUMP

Introduction to Long Jump

Long jump is a popular track and field event in athletics in which an athlete runs along a track and jumps as far as possible into a sand pit. It is one of the oldest events in athletics and is included in national and international competitions such as the Olympic Games.

The event tests an athlete's speed, strength, coordination, balance, and jumping ability. The long jump consists of four main phases: approach run, take-off, flight, and landing. The athlete aims to achieve maximum horizontal distance without crossing the take-off line.

Basic Rules of Long Jump

1. The athlete must take off from one foot only.
2. The jump is measured from the take-off line to the nearest mark made in the sand pit.
3. If the athlete crosses the take-off board line, it is considered a foul jump.
4. Each athlete is usually given three to six attempts in competitions.
5. The athlete with the longest valid jump wins the event.
6. Landing must be inside the sand pit.
7. The take-off must be completed within the designated runway area.
8. Athletes should not perform somersaults during the jump.
9. Measurement is recorded in meters and centimeters.
10. In case of a tie, the second-best jump is considered.

Court / Field Dimensions of Long Jump

Feature	Measurement
Runway Length	40–45 meters
Runway Width	1.22 meters
Take-off Board Length	1.22 meters
Take-off Board Width	20 cm
Sand Pit Length	Minimum 9 meters
Sand Pit Width	2.75–3 meters
Distance Between Board and Pit	About 1 meter

Field Description

- The athlete starts from the runway and gains speed before take-off.
- The take-off board is fixed near the sand pit.
- The landing area is filled with soft sand for safe landing.
- White boundary markings are used for measurements.
- The runway surface should be smooth and non-slippery.

Importance of Long Jump

- Improves speed, strength, and flexibility.
- Develops balance and body coordination.
- Enhances leg power and athletic performance.
- Builds concentration and confidence.
- Encourages discipline and competitive spirit.

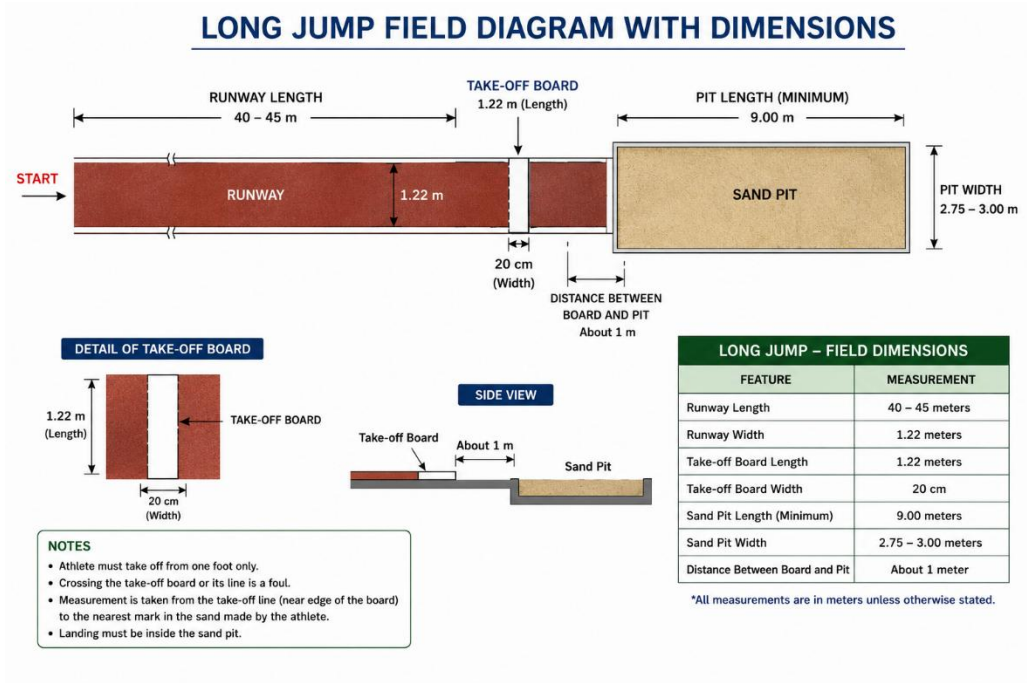


FIGURE 5.3 - LONG JUMP FIELD

Module VI

Aerobics

Introduction

Aerobics is a form of physical exercise that involves continuous and rhythmic body movements performed with the help of oxygen. The word “aerobic” means “with oxygen,” which indicates that the body uses oxygen to produce energy during exercise. Aerobics is usually performed to music and includes activities such as walking, jogging, dancing, cycling, jumping, and step exercises.

Aerobics became popular as a fitness activity because it improves the efficiency of the heart, lungs, and circulatory system. It is widely practiced in schools, gyms, sports centers, and fitness clubs. In sports, aerobics helps athletes improve stamina, flexibility, coordination, and overall physical fitness.

Key Aspects of Aerobics

1. Continuous Physical Activity

Aerobics involves continuous body movement for a longer duration without stopping. These movements increase the heart rate and improve blood circulation throughout the body.

2. Use of Oxygen

During aerobic exercise, the body uses oxygen to produce energy for muscles. This helps improve breathing efficiency and strengthens the cardiovascular system.

3. Rhythmic Movements

Aerobic exercises are usually performed in a rhythmic pattern, often along with music. This makes the activity enjoyable and helps maintain proper timing and coordination.

4. Improves Endurance

Regular aerobic exercise increases stamina and endurance, allowing athletes and individuals to perform physical activities for a longer time without getting tired quickly.

5. Whole-Body Exercise

Aerobics works on multiple body parts such as legs, arms, shoulders, and core muscles, providing complete body fitness.

Types of Aerobics

1. Low-Impact Aerobics

This type includes simple exercises where one foot always remains on the ground. It is suitable for beginners, older adults, and people recovering from injuries.

2. High-Impact Aerobics

This involves jumping and fast movements where both feet may leave the ground. It helps improve cardiovascular fitness and burns more calories.

3. Water Aerobics

Exercises are performed in water, reducing pressure on joints and muscles. It is beneficial for people with joint pain or injuries.

4. Dance Aerobics

Dance movements are combined with exercise routines and music to improve fitness while making workouts enjoyable.

5. Step Aerobics

This involves stepping up and down on a raised platform to improve leg strength, balance, and endurance.

Benefits of Aerobics in Sports

Physical Benefits

- Improves heart and lung efficiency.
- Increases stamina and endurance.
- Strengthens muscles and bones.
- Helps maintain healthy body weight.
- Improves flexibility and coordination.

Mental Benefits

- Reduces stress and anxiety.
- Improves mood and confidence.
- Enhances concentration and mental relaxation.

Sports Performance Benefits

- Helps athletes recover faster after activity.
- Improves speed and energy levels.
- Increases overall fitness required for sports competitions.

Importance of Aerobics

Aerobics is important for maintaining a healthy lifestyle and improving physical fitness. It helps prevent diseases such as obesity, high blood pressure, and heart problems. In sports training, aerobics is used to build endurance and prepare athletes for long-duration activities. Regular aerobic exercise also improves discipline, teamwork, and self-confidence.

Safety Measures During Aerobics

1. Wear comfortable sports clothing and shoes.
2. Perform warm-up exercises before starting.
3. Drink enough water to stay hydrated.
4. Avoid overexertion and exercise according to fitness level.
5. Maintain proper posture and body movements.
6. Cool down and stretch after exercise.

Conclusion

Aerobics is an effective and enjoyable form of exercise that improves both physical and mental health. It strengthens the cardiovascular system, increases stamina, and enhances overall fitness. In sports, aerobics plays an important role in improving endurance, flexibility, and athletic performance. Regular participation in aerobic activities helps individuals stay healthy, active, and energetic.



FIGURE: AEROBICS

CONCLUSION

- Physical Education plays an important role in improving physical fitness, mental health, discipline, and overall personality development
- Understanding postural deformities helps athletes prevent injuries and maintain correct body posture for better sports performance
- Stress management techniques are essential for maintaining emotional balance, improving concentration, and enhancing athletic performance.
- Throwball develops teamwork, coordination, reflexes, communication skills, and sportsmanship among players
- Long jump improves speed, strength, balance, flexibility, and confidence through proper athletic training.
- Aerobics enhances cardiovascular fitness, stamina, endurance, flexibility, and mental relaxation.
- Regular participation in sports and physical activities promotes a healthy lifestyle and reduces the risk of health problems
- Physical Education encourages discipline, leadership qualities, social interaction, and self-confidence in students.
- Overall, sports and fitness activities contribute to the holistic development of an individual and help in leading an active and healthy life.